things in common—the noble gases, the column on the far right, are all loath to chemically react, whereas the halogens, right next to them, are especially volatile. These patterns were discovered before the table was even complete, and in fact the creator of the table, Dmitri Mendeleev, left gaps for elements he knew *should* exist based on the pattern, even before they were discovered.

The patterns in the periodic table led to theorizing about electron orbitals, which led to discoveries about the fundamental nature of subatomic matter. Over and over again, scientists have developed new theories of nature by recognizing patterns in their observations and then looking for a hidden property that could give them insight into what was really going on. We do this all the time, ourselves, without noticing it. Watching highway traffic change over the course of the day can tell you standard business hours. The pattern of fading in a carpet can let you deduce which parts of the room get the most sunlight (and thereby, indirectly, tell you how the Earth and Sun are oriented in the Solar System).

In the case of particle physics, using symmetry is often a lot like building new periodic tables, but for even smaller building blocks of nature. Similarities between particles—their charge, mass, or spin, for instance—can give clues about similarities in their formation or their connections to fundamental forces. Arranging these particles by their patterns lets physicists identify the symmetries that can be the defining features of entire theories.

Sometimes those patterns are most easily seen mathematically. If you write down an equation to describe a physical process, and then find that you can swap some terms around without actually changing the physical phenomenon the equation describes, you've found a mathematical symmetry. And it's probably telling you something deep about the particles or fields you're describing.

This symmetry-oriented way of looking at particles and the relationships between them is so prevalent in physics that we find ourselves using references to mathematical symmetries as shorthand for the theories themselves. For instance, electromagnetism is frequently called U(1) theory, because some aspects of the mathematics have the same kind of symmetry as a circle, and "U(1)" is shorthand for a mathematical group that describes rotations around a circle.